1 – Introduction & Basic Concepts

- $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{sys}}$
  First Law – energy conservation for a system.
- $E_{\text{total}} = U + KE + PE$, $KE = \frac{1}{2}mV^2$, $PE = mgz$
  Total energy includes internal, kinetic and potential components.
- $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}$, $P = P_0 + \rho g h$
  Absolute pressure relation and hydrostatic equation.
- $T(K) = T(^\circ C) + 273.15$, $T(^\circ F) = 1.8T(^\circ C) + 32$
  Temperature scale conversions.
- $\rho = m/V$, $v = 1/\rho$, $\gamma = \rho g$
  Density, specific volume and specific weight.
- $p_{\text{gage}} = \rho g h$, $\Delta p = (\rho_2 - \rho_1) g h$
  Manometer and differential pressure relations.

2 – Energy & General Analysis

- $\Delta E = Q - W$
  First Law for a closed system (energy balance).
- $\Delta U = Q - W_b - W_{\text{other}}$, $W_b = \int P\,dV$
  Internal energy change including boundary work and other work modes.
- $\eta = \dfrac{\text{useful output}}{\text{energy input}}$
  General efficiency definition.
- $w_{\text{flow}} = Pv$, $e_{\text{mech}} = Pv + \frac{V^2}{2} + gz$
  Flow work and specific mechanical energy for flowing fluids.
- $e_{\text{flow}} = h + \frac{V^2}{2} + gz$, $\dot{E}_m = \dot{m}(h + \frac{V^2}{2} + gz)$
  Energy transport with mass flow in open systems.

3 – Properties of Pure Substances

- $x = \frac{m_v}{m_t}$, $y = y_f + x(y_g - y_f)$
  Quality and mixture property interpolation.
- $h = u + Pv$
  Definition of enthalpy.
- $Pv = RT$, $Z = Pv/(RT)$
  Ideal gas law and compressibility factor.
- **State Principle:** two independent intensive props fix the state.
- **Specific vs Total:** per mass properties (v, h, u, s) vs totals (V, H, U, S).

4 – Energy Analysis of Closed Systems

- $W_b = \int_{V_1}^{V_2} P\,dV$
  General boundary work.
- $W_b = P(V_2 - V_1)$
  Isobaric process work.
- $W_b = P_1 V_1 \ln(V_2/V_1)$
  Isothermal ideal gas work.
- $W_b = \frac{P_2V_2 - P_1V_1}{1-n}$, $n \neq 1$
  Polytropic process work.
- $W_b = \frac{P_1 + P_2}{2}(V_2 - V_1)$
  Linear p–v relation (trapezoidal area).
- $Q - W = \Delta U + \Delta KE + \Delta PE$
  Complete energy balance for a closed system.
- $c_v = (\partial u/\partial T)_V$, $c_p = (\partial h/\partial T)_P$
  Definitions of specific heats.
- $\Delta u \approx c_v(T_2 - T_1)$, $\Delta h \approx c_p(T_2 - T_1)$
  Constant–specific–heat approximation for ideal gases.
- $c_p - c_v = R$
  Mayer's relation for ideal gases.

5 – Control Volumes (Mass & Energy)

- $\sum m_{\text{in}} - \sum m_{\text{out}} = \dfrac{dm_{\text{sys}}}{dt}$
  Mass conservation for control volumes.
- $\dfrac{dE_{cv}}{dt} = \sum \dot{Q}_{in} - \sum \dot{Q}_{out} + \sum \dot{m}\left(h + \frac{V^2}{2} + gz\right)_{in} - \sum \dot{m}\left(h + \frac{V^2}{2} + gz\right)_{out}$
  General energy rate balance for open systems.
- **Steady–Flow Devices:**
  - Nozzle/Diffuser: $h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$
    Converts enthalpy kinetic energy (adiabatic).
  - Turbine: $w_t = h_1 - h_2$
    Work output from enthalpy drop.
  - Compressor/Pump: $w_c = h_2 - h_1$
    Work input to raise enthalpy or pressure.
  - Throttle: $h_2 \approx h_1$
    Isenthalpic pressure drop (no work).
  - Mixing chamber: $\sum \dot{m}_{in} h_{in} = \sum \dot{m}_{out} h_{out}$
    Adiabatic mixing, no shaft work.
  - Heat exchanger: $\dot{Q}_{hot} = -\dot{Q}_{cold}$
    Energy transfer between streams (no work).

6 – Thermodynamic Property Relations

- $H = U + PV$, $h = u + Pv$
  Definition of enthalpy (total and specific).
- $Tds = du + Pdv$, $Tds = dh - vdP$
  Tds relations linking state properties.
- $ds = \dfrac{du}{T} + \dfrac{Pdv}{T}$, $ds = \dfrac{dh}{T} - \dfrac{vdP}{T}$
  Differential forms for calculating entropy changes.
- $R = C_p - C_v$, $k = C_p/C_v$
  Relations between specific heats and gas constant.
- Incompressible: $ds = c\,\dfrac{dT}{T}$, $s_2 - s_1 = c_{avg}\ln(T_2/T_1)$
  Entropy change for liquids/solids (constant c).
- $dh = C_p\,dT$, $du = C_v\,dT$
  Constant–property relations for ideal gases.
- $\Delta h = C_p(T_2 - T_1)$, $\Delta u = C_v(T_2 - T_1)$
  Ideal–gas property change formulas.
- $\Delta s = C_v \ln\frac{T_2}{T_1} + R\ln\frac{v_2}{v_1}$
  Entropy change in terms of T and v.
- $\Delta s = C_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}$
  Entropy change in terms of T and P.
- $h = u + Pv$, $du = Tds - Pdv$, $dh = Tds + vdP$
  Combined property differentials for analysis of reversible paths.

7 – Entropy Fundamentals

- $\oint \frac{\delta Q}{T} \leq 0$
  Clausius inequality — equality for reversible cycles.
- $dS = \left(\frac{\delta Q}{T}\right)_{\text{rev}}$
  Definition of differential entropy change.
- $\Delta S = \int_1^2 \frac{\delta Q_{\text{rev}}}{T}$
  Entropy change between two states.
- $\Delta S_{\text{isoth}} = \frac{Q}{T_0}$
  Entropy change for reversible isothermal process.
- $\Delta S_{\text{sys}} = \int \frac{\delta Q}{T} + S_{\text{gen}}$
  General form with entropy generation ($S_{gen} \geq 0$).
- $\Delta S_{\text{isol}} \geq 0$
  Entropy of isolated systems never decreases.
- $Tds = dU + PdV$, $Tds = dh - vdP$
  Combined 1st + 2nd law differentials.
- $S = k\ln W$, $S = -k\sum_i p_i \ln p_i$
  Statistical definition of entropy.

8 – Isentropic & Reversible Processes

- $PV^k = \text{const}$ (for ideal gas)
  Isentropic relation for reversible adiabatic processes.
- $T_2/T_1 = (P_2/P_1)^{(k-1)/k}$
  Temperature–pressure relation, isentropic ideal gas.
- $T_2/T_1 = (v_1/v_2)^{k-1}$
  Temperature–volume relation, isentropic ideal gas.
- $P_2/P_1 = (v_1/v_2)^k$
  Pressure–volume form of isentropic law.
- $w = \frac{P_2v_2 - P_1v_1}{1-k}$
  Work done in an isentropic process ($k \neq 1$).
- $q_{\text{rev}} = \int T\,ds$
  Heat transfer for reversible process.

9 – Gas Power Cycles

- $\eta = 1 - \dfrac{Q_{\text{out}}}{Q_{\text{in}}}$
  Thermal efficiency of any cycle.
- $W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}}$
  Net work per cycle.
- **Otto:** $\eta = 1 - \dfrac{1}{r^{k-1}}$, $r = V_1/V_2$
  Efficiency for spark-ignition (constant-volume) engine.
- **Diesel:** $\eta = 1 - \dfrac{1}{r^{k-1}}\dfrac{(\rho^k - 1)}{k(\rho - 1)}$
  Efficiency for compression-ignition (constant-pressure) engine.
- **Brayton:** $\eta = 1 - \dfrac{1}{r_p^{(k-1)/k}}$, $r_p = P_2/P_1$
  Ideal gas-turbine efficiency.
- $W_{\text{net}} = c_p[(T_3 - T_4) - (T_2 - T_1)]$
  Net work per unit mass for Brayton cycle.
- $\varepsilon = (T_5 - T_2)/(T_4 - T_2)$
  Regenerator effectiveness.
- $\eta_c = \dfrac{h_{2s} - h_1}{h_2 - h_1}$, $\eta_t = \dfrac{h_3 - h_4}{h_3 - h_{4s}}$
  Isentropic efficiencies of compressor and turbine.

10 – Vapour Power Cycles (Rankine)

- $\eta = 1 - \dfrac{h_4 - h_1}{h_3 - h_2}$
  Thermal efficiency of Rankine cycle.
- $w_t = h_3 - h_4$, $w_p = h_2 - h_1$
  Turbine output and pump input per kg.
- $q_{\text{in}} = h_3 - h_2$, $q_{\text{out}} = h_4 - h_1$
  Boiler and condenser heat transfers.
- $w_{\text{net}} = (h_3 - h_4) - (h_2 - h_1)$
  Net cycle work.
- **Reheat:** $w_t = (h_3 - h_4) + (h_5 - h_6)$
  Work for two-stage expansion with reheat.
- **Regeneration:** $y = \dfrac{h_f - h_2}{h_y - h_2}$
  Extraction (bleed) steam fraction for open feedwater heater.
- $\text{UF} = \dfrac{W_{\text{net}} + Q_{\text{proc}}}{Q_{\text{in}}}$
  Utilisation factor for cogeneration.

11 – Refrigeration & Heat Pump Cycles

- $\text{COP}_R = Q_L/W_{\text{in}}$, $\text{COP}_{HP} = Q_H/W_{\text{in}} = 1 + \text{COP}_R$
  Definitions of coefficient of performance.
- $\text{COP}_{R,C} = \dfrac{T_L}{T_H - T_L}$
  Carnot refrigeration COP.
- $\text{COP}_R = \dfrac{h_1 - h_4}{h_2 - h_1}$
  Vapour-compression COP (actual).
- $w_c = h_2 - h_1$, $q_L = h_1 - h_4$, $q_H = h_2 - h_3$
  Compressor work and heat transfers.
- $\eta_c = \dfrac{h_{2s} - h_1}{h_2 - h_1}$
  Isentropic compressor efficiency.
- $\text{COP}_R = 1/(r_p^{(k-1)/k} - 1)$
  Gas-refrigeration (Brayton) COP.

12 – Isentropic Devices & Analysis

- $\Delta h = c_p(T_2 - T_1)$, $\Delta u = c_v(T_2 - T_1)$
  Constant specific heat approximations for ideal gases.
- $\Delta h = \int c_p(T)\,dT$, $\Delta u = \int c_v(T)\,dT$
  Variable specific heat integration forms.
- $\Delta s = \int \frac{c_p(T)}{T}\,dT - R\ln\frac{p_2}{p_1}$
  Entropy change for ideal gas with variable c_p.
- $\frac{T_2}{T_1} = (\frac{p_2}{p_1})^{(k-1)/k}$, $\frac{p_2}{p_1} = (\frac{v_1}{v_2})^k$, $\frac{T_2}{T_1} = (\frac{v_1}{v_2})^{k-1}$
  Isentropic relations for ideal gas with constant k.
- $p_2/p_1 = P_{r2}/P_{r1}$, $v_2/v_1 = V_{r2}/V_{r1}$
  Relative pressure/volume for variable c_p (use Table A-17).
- $Q - W = \Delta U + \Delta KE + \Delta PE$ (closed)
  First law for closed systems.
- $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)]$ (open)
  First law for steady-flow devices.
- $\dot{W}_t = \dot{m}(h_1 - h_2)$
  Turbine work output.
- $\dot{W}_c = \dot{m}(h_2 - h_1) \approx \dot{m}v\Delta p$
  Compressor/pump work input.
- $\eta_t = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$, $\eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}$, $\eta_p = \frac{v_1(p_2 - p_1)}{h_{2a} - h_1}$
  Isentropic efficiencies for turbine, compressor, and pump.

13 – Gas Power Cycles

- $\eta = 1 - \frac{Q_L}{Q_H} = W_{net}/Q_H$
  Thermal efficiency for any heat engine cycle.
- $\text{COP}_R = \frac{Q_L}{W_{net}}$, $\text{COP}_{HP} = \frac{Q_H}{W_{net}}$
  Coefficient of performance for refrigerators and heat pumps.
- $V_D = V_{max} - V_{min}$, $r = \frac{V_{BDC}}{V_{TDC}}$, $p_{mep} = W_{net}/V_D$
  Displacement volume, compression ratio, and mean effective pressure.
- **Otto:** $\eta = 1 - \frac{1}{r^{k-1}}$, $w_{net} = c_v[(T_3 - T_4) - (T_2 - T_1)]$
  Spark-ignition cycle efficiency and net work (1–2 isent. comp., 2–3 const-V heat add, 3–4 isent. exp., 4–1 const-V reject).
- **Diesel:** $\eta = 1 - \frac{1}{r^{k-1}}\frac{\beta^k - 1}{k(\beta - 1)}$, $\beta = V_3/V_2$
  Compression-ignition cycle efficiency and cut-off ratio (2–3 const-P heat add).
- $T_2 = T_1 r^{k-1}$, $T_3 = T_2\beta$, $T_4 = T_3(\beta/r)^{k-1}$
  Temperature relations for Diesel cycle.
- **Stirling/Ericsson:** $\eta = 1 - \frac{T_L}{T_H}$
  Carnot efficiency for constant-temperature heat addition/rejection cycles.
- **Brayton:** $\eta = 1 - \frac{1}{r_p^{(k-1)/k}}$, $r_p = p_{max}/p_{min}$
  Gas-turbine cycle efficiency and pressure ratio (1–2 isent. comp., 2–3 const-P heat add, 3–4 isent. exp., 4–1 const-P reject).
- $T_{2a} = T_1 + \frac{T_{2s} - T_1}{\eta_c}$, $T_{4a} = T_3 - \eta_t(T_3 - T_{4s})$
  Non-ideal Brayton temperatures with compressor/turbine efficiencies.
- $\varepsilon = \frac{h_5 - h_2}{h_4 - h_2}$ (regeneration), $r_{stage} = \sqrt{r_p}$ (intercooling)
  Regenerator effectiveness and optimal stage pressure ratio for Brayton improvements.
- $R_{BW} = w_c/w_t \approx 0.4$–$0.6$
  Back work ratio for gas-turbine cycles.
- **Rankine:** $w_p = v(p_2 - p_1)$, $w_t = h_3 - h_4$, $\eta = 1 - \frac{q_{out}}{q_{in}}$
  Steam cycle: pump work, turbine work, and efficiency (1–2 pump, 2–3 boiler, 3–4 turbine, 4–1 condenser).
- **Reheat:** $q_{in} = (h_3 - h_2) + (h_5 - h_4)$, $w_t = (h_3 - h_4) + (h_5 - h_6)$
  Two-turbine Rankine with reheater; improves efficiency and steam quality.
- $\eta_p = \frac{v(p_2 - p_1)}{h_{2a} - h_1}$, $\eta_t = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$
  Pump and turbine isentropic efficiencies for real Rankine analysis.